MILITARY SPECIFICATION

MIL-F-55006(SigC)
9 January 1958

FACSIMILE SET AN/TXC-1( )

1. SCOPE

1.1 This specification covers one type of equipment designated as Facsimile Set AN/TXC-1( ). The AN/TXC-1( ) consists of Facsimile Transceiver TT-I( )/TXC-1, Rectifier Power Unit PP-86( )/TXC-1, Table MT-252, ( )/TXC-1 Exhaust Filter Assembly, and Loudspeaker LS-11( ). They are used to transmit and receive written, printed, or pictorial matter over wire or short radio communication facilities.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-B-36	Bags, Paper (Grocers)
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing
UU-T-111	Tape, Paper, Gummed, Water-Resistant
UT-276	Thread cotton
LLL-F-291	Fiberboard; Corrugated, Single-Face(Flexible)
CCC-C-419	Cloth, Cotton Duck
QQ-5-571	Solder, Soft Lead Alloy, Tin Lead Alloy, Flux Cored Ribbon
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
T-T-871	Twine, Cotton, Wrapping
MS-20003	Indicator, Humidity, Card, Three Spot Impregnated Areas

MILITARY

MIL-E-1	Army-Navy Preferred List of Radio Electron Tubes
MIL-P-116	Packaging and Packing for Overseas Shipment
MIL-B-117	Preservation, Methods of
JAN-P-125	Packaging and Packing for Overseas Shipment
MIL-B-131	Barrier-Material, Waterproof, Flexible
MIL-T-152	Treatment, Moisture-and-Fungus-Resistant of Communications, Electronic, and Associated Electrical Equipment, General Process For

FSC 5815

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MIL-D-504	Dyeing and Aftertreating Processes
JAN-P-642	Plug PL-68 (Three Conductor Microphone)
MIL-C-9074	Cloth, Sateen, 2 Ply
MIL-M-13231	Identification Marking of Signal Equipment
MIL-F-14072	Finishes
MIL-E-15090	Enamel, Equipment, Light-Grey (Formula No. 111)
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, and Nickel Alloys

STANDARDS

MILITARY

MIL-STD-105	Humidity Test Cycle for Ground Signal Equipment
MIL-STD-129	Marking for Shipments and Storage
MIL-STD-170	Moisture Resistance Test for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects for

DRAWINGS

SIGNAL CORPS

SC-D-339	Plug PL-55
SC-D-375	Plug PL-68
SC-D-4821	Cord CD-1018, Assembly and Details
SC-D-4822	Cord CD-1019, Assembly and Details
SC-DL-36963	TT-1( )/TXC-1
SC-DL-37426	PP-86( )/TXC-1
SC-DL-37476	MT-252( )/TXC-1
SM-D-201304	Drum
SC-DL-201677	Exhaust Filter Assembly
SC-C-33073	Form to be Used for Pilot Pack Process Sheet

2.2 Other publications.-

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, 1424 16th Street, N.W., Washington,6, D. C.)

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification

(Application for copies should be addressed to the Association of American Railroads, 202 Chicago Union Station, Chicago 6, Illinois)

GPO 810020-1

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3. REQUIREMENTS

3.1 Construction.- Facsimile Transceiver shall be constructed in accordance with Drawing SC-DL-36963 and the requirements of this specification. In case of conflict between the specification and drawings, the specification shall govern.

3.2 Brazing, soldering, welding and cleaning.-

3.2.1 Brazing.- Brazing of steel, copper, copper alloy, and nickel alloys shall conform to Specification MIL-B-7883.

3.2.2 Soldering.- The solder used for electrical connections shall be Stock Number 60 conforming to Specification QQ-S-571.

3.2.2.1 Flux and cleaning agents.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.2.2.2 Process.- There shall be no sharp or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be over-heated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not include the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment, flux being spread over a larger area, or resulting damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.2.3 Welding.- Welds shall be of ample size. Where tack or spot welds are employed, the number of welds shall be sufficient to provide adequate strength and at least two welds shall be used to hold each welded part. Spot or tack welds in aluminum or magnesium shall be limited to lightly stressed, non-critical structures. Wherever possible, the welded joint shall be such that grinding on the finished weld will be unnecessary.

3.2.3.1 Cleaning surfaces before welding.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.2.3.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, and other harmful defects. Where undesirable internal stresses are likely to result from welding, the welded assembly shall be stress-relieved. Insert-gas-shielded arc welding shall be used when practicable for welding of aluminum or magnesium.

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3.2.4 Cleaning.- Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no visible or latent deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly, components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.3 Finish, protective.- The equipment shall be given protective finish in accordance with Specification MIL-F-14072. The final paint film on type I surfaces shall be semigloss, light-gray enamel conforming to Specification MIL-E-15090. The aluminum drum and the stainless steel lead screw shall not be finished.

3.4 Marking.-

3.4.1 General.- Marking shall conform to Specification MIL-M-13231.

3.5 Tropicalization.-

3.5.1 Treatment of assembled equipment.- Assembled equipment shall be treated in accordance with Specification MIL-T-152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc.) which is applicable on contract.

(2) It does not incorporate any material listed in 3.5.3 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of Specification MIL-F-14072.

3.5.2 Treatment of materials.-

3.5.2.1 Treating materials.- Treating materials containing a mercury bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.5.2.2 Toxicity.- Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.5.2.3 Flexibility.- Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

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3.5.2.4 Statement of treatment.- The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with Specification MIL-T-152, the above statement shall be incorporated in the statement of treatment required by that specification.

3.5.3 Materials not resistant to moisture and fungi.- The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.
(b) Cotton.
(c) Cork.
(d) Felt, hair or wool.
(e) Jute.
(f) Leather.
(g) Linen.
(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.
(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.6 Covers.- The dust covers shall be provided for the transceiver and rectifier power unit to protect the equipment when not in operation and shall be constructed in accordance with the models and materials as follows:

a. The covers on the TT-1( )/TXC-1 and PP-86( )/TXC-1 shall be made of cotton duck #10, Type I in accordance with Specification CCC-C-419. The thread shall be in accordance with Specification V-T-276, color shade 7 O.D. The covers shall be treated in accordance with Specification MIL-D-504, Class B, after construction.

3.7 Mounting.- Facsimile Transceiver TT-1( )/TXC-1 and Rectifier Power Unit PP-86( )/TXC-1 shall mount on Table MT-252( )/TXC-1.

3.8 Power Source.- The transceiver shall be capable of operation on power delivered by Rectifier Power Unit PP-86( )/TXC-1.

3.9 E-2b. Frequency Band.- The signal employed shall consist of an 1800 cycle-per-second carrier frequency, amplitude modulated. The modulation frequency employed shall not exceed 900 cycles-per-second.

3.10 Copy Size.- The drum of the transceiver shall be capable of accepting sheets 12 inches by 18-11/16 inches, plus zero, minus 1/16 inch, and shall be capable of reproducing an area of at least 12 inches by 18 inches. The nominal diameter of the drum shall be 6 inches. (See Drawing SM-D-201304 (SigC)

3.11 Picture Definition.- The transceiver shall have a resolving power of at least 96 lines per inch along both X-axis and Y-axis of the picture. There shall be 96 scanning lines per inch.

3.12 Cylinder Speed.- The cylinder speed shall be maintained as closely as possible to 60 rpm under control of the frequency generator specified in 3.13 and 3.14.

3.13 Synchronization.- Each of two transceivers forming a circuit shall be held in synchronism by an 1800 cycle per second frequency generator contained in each equipment. Synchronization shall not be transmitted from one unit to the other during the transmission of the copy.

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3.14 Frequency Stability.- The frequency of the synchronizing signal shall be adjustable to one part in 324,000 at any ambient temperature between 0 degrees and 49 degrees C. The temperature coefficient of the frequency generator and its associated amplifier shall be at least one part in 3,470,000 per degree C with the extreme limits of 0 degrees and 49 degrees C. This is equivalent to no more than a 0.070 inch skew (uniform) in a 12 inch picture received on an equipment operating in an ambient temperature plus or minus 11 degrees Centigrade from the ambient temperature of the transmitter, provided the above ambient temperature falls between the limits of 0 degrees Centigrade and 49 degrees Centigrade, and the fork frequency of the receiver has been adjusted to zero skew when the ambient temperature of the transmitter and the receiver are the same.

3.15 Phasing Signals.- Phasing signals shall be generated by scanning a phasing ring. The phasing ring shall be a dull black ring or equivalent with a white insert 1/4 inch long as measured on the circumference of the ring. The phasing ring shall be located at the left end of the drum. The phasing pulse shall increase or decrease the steady carrier signal by at least 8 db for positive transmission, and by at least 6 db for negative transmission. The drum shall be capable of being moved to the right far enough to scan the entire phasing ring.

3.16 Color Response.- The transceiver shall be capable of transmitting and receiving photographs, black-and-white copy, or colored copy such as maps of the U.S. Coast and Geodetic Survey, weather maps, or marks made on white paper by any of the following colored pencils:

Green	Venus No. 1208
Blue	Venus No. 1206
Yellow	Venus No. 1229
Red	Eagle No. 745
Orange	Eagle No. 737

It need not be possible to distinguish between the various colors in the received copy.

3.17 Recording.- The transceiver shall be capable of recording on any of the following mediums:

(1) Transmission Film, Type A, as made by Eastman Kodak Co., or equal.

(2) Royal Bromide Paper, F-1, as made by Eastman Kodak Co., or equal.

(3) Teledeltos, Grand H, as made by Western Union Telegraph Co., or equal.

(4) Timefax, as supplied by Times Telephoto Equipment Inc., or equal.

3.18 Transmission.- It shall be possible to transmit in a positive or negative sense. A transmission in the positive sense is defined as one in which the darker portions of the original are converted into maximum signals and the lighter portions of the original are converted into minimum signals. A negative transmission is defined as the reverse, the lighter portions giving maximum and the darker portions minimum signals. In either case, it shall be possible to vary the difference between maximum and minimum signals from a minimum of 6 db to at least 20 db.

3.19 Power Level Indicator.- A meter shall be mounted on the transceiver to indicate the relative signal level during transmission and reception.

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3.20 Termination.- Suitable input and output facilities shall be provided in the transceiver for matching the input and output impedances and levels of the transceiver to any of the input and output impedances specified below.

3.20.1 Transmitting.- The output of the transceiver shall be matched as follows:

Load Impedance, Ohms	Termination	Output Level with Meter Reading 2db
600	Miniature female connector one side grounded	+26 dbm across 600 ohms
600	3 binding posts, center tapped	0 dbm across 600 ohms
100	2 jacks in parallel * to fit Plug PL-68	0.4 volt across 100 ohms

*Provision shall be made for automatically shorting tip to sleeve of Plug PL-68 when the selector switch is turned from Set Range to Transit.

3.20.2 Receiving.- The input of the transceiver shall be matched as follows:

Load Input Ohms	Termination	Minimum Input Operating Level
600	Miniature female connector, one side grounded	-45 dbm
2500	2 jacks in parallel to fit Plug PL-55	0.25 V across 2500 ohms
600	3 binding posts, center-tapped, centertap ungrounded	-43 dbm

3.21 Coupling to Telephone Instrument.- When the transceiver is inductively coupled to a transmission line by means of a telephone receiver and Type UC coupling coil, the resulting signal on the transmission line shall be approximately -8 dbm.

*The UC coupling coil shall provide two outputs with nominal impedance of 3 ohms and 15 ohms. When the plug of the miniature connector is connected to Facsimile Set AN/TXC-1( ) and the voltage across the miniature connector is adjusted to 15.46 volts, the unloaded voltage across the 30- ohm winding shall be 0.78 volt nominal and the output across the 15-ohm winding shall be 6.3 volts nominal.

3.22 Tubes, Electron.- The following sets of tubes shall be provided with each transceiver in accordance with Specification MIL-E-1.

3.22.1 Electron Tubes in Use.-

Quantity	Type
3	1635
4	7C5
1	7C7
4	7L7
3	7N7
1	1645
1	884
1	R1130B
1	5651

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3.23 Cords CD-1018 and CD-1019.- Cords CD-1018 and CD-1019 shall be in accordance with Drawings SC-D-4821 and SC-D-4822, respectively.

3.24 Talkback Circuit.- A talkback circuit shall be provided in Facsimile Transceiver TT-1( )/TXC-1 for the purpose of providing voice communications while the circuit is not being used for facsimile signals.

3.25 Pilot Lamp.- The operating voltage across the lamp shall be approximately 3.15 volts.

3.26 Noise.- Acoustic noise from the transceiver shall be kept to a minimum.

3.27 Stylus.- The nominal diameter of the stylus shall be 0.008 inch.

3.28 Motor Current.- The motor current shall be set between 60 and 75 milliamperes.

3.29 Motor.- Facsimile Transceiver TT-1( )/TXC-1 shall contain a synchronous motor to operate the drum. It shall be Times Facsimile Corporation type KBA-2, or equal, and shall conform to the dimensions shown on Drawing SM-D-200975. The motor shall include a primary and secondary clutch assembly and a phasing magnet assembly.

3.30 Tuning Fork Oscillator.- Facsimile Transceiver TT-1( )/TXC-1 shall contain a tuning fork oscillator to furnish a source of 1800 cycle voltage for the sub-carrier and for the motor amplifier. The output frequency of the tuning fork oscillator shall be controlled by a metallic tuning fork enclosed in a metal case.

3.30.1 Tuning Fork Case.- The tuning fork case shall be dehydrated, evacuated and sealed. No coils, varnish or other materials capable of absorbing a large volume of gas or evaporating under low pressure shall be placed within the tuning fork case. The tuning fork case, driving coils and leads shall be installed in an outer case which shall then be packed with shredded cork. The outer case shall then be hermetically sealed. The tuning fork shall be sufficiently balanced to permit operation while tilted at an angle of 45°.

3.30.2 Stability of Tuning Fork Oscillator.- The output frequency of the tuning fork oscillator shall not vary when the atmospheric pressure is varied from the normal atmospheric pressure at sea level to the atmospheric pressure at 10,000 feet. After prolonged storage at atmospheric pressure occurring at any altitude up to 50,000 feet, the nominal frequency of the tuning fork oscillator shall not differ from the frequency measured before storage as a result of leakage into the tuning fork case.

3.30.3 Oscillator Output.- The tuning fork oscillator shall furnish voltage to the modulator of Facsimile Transceiver TT-1( )/TXC-1. Another output circuit shall furnish voltage to the motor amplifier. The output voltage and impedance shall be as follows:

Circuit	Output Voltage	Nominal Impedance
Modulator Circuit	0.72 volt	470 ohms
Motor Amplifier	38 volts	20000 ohms

The output for the modulator shall have a total harmonic content at least 30 db lower than the voltage level of the fundamental frequency. (1800 cps.)

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3.31 Input Circuit.- Input circuit shall be designed to discriminate against low frequency noise or hum. The gain at 60 cycles shall be 26 db below the gain at 1800 cps. The input circuit shall contain an input transformer with a balanced primary winding. The balance between each half shall be within 5 per cent.

3.32 Lens Mountings.- Wherever glass surfaces of lenses are assembled in metal framework, lead washers shall be inserted between the glass surfaces and the metal framework.

3.33 Alternate Speed.- Means shall be provided within Transceiver TT-1( )/TXC-1 to reduce the frequency of the synchronous motor supply to 1000 cps. This 1000 cycle source shall be derived from the 1800 cycle tuning fork oscillator. The frequency dividing circuit shall retain the frequency stability of the tuning fork oscillator and shall contain no free running milti-vibrators or other oscillator circuits. A suitable switch shall be provided to select either 1800 cps or 1000 cps for the motor amplifier circuit. This switch shall be mounted on the chassis inside the set and shall be readily available upon removal of the back cover.

3.34 Rectifier Power Unit PP-86( )/TXC-1.-

3.34.1 Function.- The rectifier power unit shall supply plate and filament power to Facsimile Transceiver TT-1( )/TXC-1. The power unit shall contain a constant output amplifier of adequate size to supply the exciter lamp in the transceiver with power at 6 volts, 1800-cycles.

3.34.2 Input.- The rectifier power unit shall satisfactorily operate Facsimile Transceiver TT-1( )/TXC-1 with an input varying from 100 to 130 volts a-c, at any nominal frequency between 50 and 65 cycles per second and varying as much as plus or minus 5 cycles in one second.

3.34.3 Tubes, Electron.- The following set of tubes in accordance with Specification MIL-E-1 shall be provided with each rectifier power unit.

3.34.3.1 Tubes in Use.-

Quantity	Type
1	5Z3
2	7C5
1	7L7
1	7N7
1	7S7
1	5651

3.35 Transmission Test.- A test shall be made over each of the following test circuits, and shall result in the satisfactory reproduction of the transmitted copy.

(a) A wire circuit consisting of 15 miles of Signal Corps Wire WD-1/TT, or equivalent.

(b) A short radio circuit.

3.35.1 Transmission.- A transceiver shall be set up for operation with the rectifier power unit, and transmission shall be made under the conditions of input voltage and frequency variations specified in paragraph 4.8. The copy transmitted during test shall be received on photographic film on another transceiver operated at normal voltage and frequency. The received copy shall show no variations due to changes in operating frequency or voltage.

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3.35.2 Reception.- Each facsimile transceiver shall be set up for reception on film. Copy shall be transmitted from another transceiver operating at normal voltage and frequency, to the one under test. Reception shall be made while the machine under test is subjected to the voltage and frequency variations given in paragraphs 4.8.2 and 4.8.2.1. The received copy shall show no variation due to changes in frequency or voltage.

3.36 Synchronizing Frequency.- When measured as indicated in paragraph 4.9, the beat between the standard frequency and the unit under test shall not exceed one beat in three minutes.

3.37 Recording.-

3.37.1 Photographic.- The received copy at all levels shall be without faults, with an indication of +2db on the meter, the current through Tube R1130B shall be 34±1 ma., when measured with a d-c milliammeter. See paragraph 4.10.1.

3.37.2 Direct Recording.- The received copy at all levels shall be without faults. See paragraph 4.10.2.

3.37.3 Operation.- All transmission and reception shall be made to or from an approved sample transceiver. Copy provided by each transceiver being tested shall be examined for defects and shall compare favorably with copy provided by the two approved samples operating together. See paragraph 4.10.3.

3.37.4 Spot size.- The copy received in accordance with paragraph 4.10.3 shall be examined for proper size. The shape of the spot shall be square with sharply defined corners. The width of the spot shall be such that the scanning lines on a received photographic copy are wider than 0.0094 inch but not wider than 0.0114 inch. See paragraph 4.10.4.

3.37.5 Microphonic Tubes.-

3.37.5.1 Photographic copy received on Royal Bromide Paper, F-1, or equal, in accordance with paragraph 4.10.3 shall show no effects of microphonics.

3.37.5.2 The stop over mechanism shall stop and remain stationary when tested as in paragraph 4.10.5.1.

3.38 Phasing.-

3.38.1 Positive.- The transceiver shall be capable of phasing when the maximum level of the phasing signal is at each of the following readings on the transceiver meter: +4, +2, 0 and -1 dbm. See paragraph 4.11.1.

3.38.2 Negative.- The transceiver shall phase when the minimum level of the phasing signal is at each of the following readings on the transceiver meter: -10, -8, -6 and -4 dbm. See paragraph 4.11.2.

3.38.3 Motor running.- This secondary phasing operation shall require at least five but not more than seven revolutions of the lead screw and drum assembly. See paragraph 4.11.3.

3.39 Pick-Up from Local Frequency Generator.- The transceiver meter shall give no indication of signal when tested as in paragraph 4.12.

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3.40 Fork Beat.- There shall be not more than ½ db beat as indicated on the meter of the transceiver or an external meter, at the discretion of the inspector. See paragraph 4.13

3.41 Receiver Sensitivity.-

3.41.1 600 ohm Termination.- The transceiver being tested shall be capable of producing a + 2db meter reading from the test signal received through the attenuator pad when the loss in the pad is adjusted to 45 db, and also when the loss in the pad is adjusted to 15 db. See paragraph 4.14.1.

3.41.2 2500 ohm Termination.- The transceiver shall be capable of producing a + 2db meter reading in Record Photo position. See paragraph 4.14.2.

3.42 Output Level Test.- The output level shall fall within the given limits as follows:

Output	Output Level
Miniature Connector	+26 dbm ± 2 db
Binding Post	0.63 to 1.00 V.
Transmit Jack	0.3 to 0.5 V.
UC Coil (3 ohm)	0.690 to 0.869 V.
UC Coil (15 ohm)	4.89 to 7.74 V.

See paragraph 4.15

3.43 Contrast.- When the wedge is adjusted with the light focused on the white and black of the phasing ring, it shall be possible to obtain a contrast range of at least 20 db using either positive or negative transmission. It shall also be possible to reduce the contrast range to at least 6 db for either positive or negative transmission. See paragraph 4.16.

3.44 Regulation.-

3.44.1 With input at 130V a-c the change in meter readings shall not exceed 0.5 db when tested as in paragraph 4.17e.

3.44.2 With input at 100V a-c the meter reading shall not differ by more than 1 db from the reading at 130 volts a-c (RMS). See paragraph 4.17b.

3.44.3 At no time shall the reading of the transceiver meter change more than 1 db when varied as indicated in paragraph 4.17g.

3.45 Motor Test.-

3.45.1 The time required for the motor to accelerate from standstill to a speed of greater than 60 r.p.m. shall be not more than 5 seconds. See paragraph 4.18.

3.45.2 Motor Current.- Motor current should be 70 ma. ± 1 ma. when tested as in paragraph 4.18.2.

3.45.3 Motor Coasting.- Drum should stop rotating within 30 revolutions after key switch is released as in paragraph 4.18.2.

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3.45.4 Adjustment of Clutch Tension.- The primary clutch tension, when clutch is slipping, shall be between 12 and 14 ounces, with a torque arm of 3 inches. The secondary clutch tension, when clutch is slipping, shall be between 3 and 6 ounces, with a torque arm of 3 inches. See paragraph 4.18.4.

3.46 Talkback Circuit.- When using the talkback circuit, it shall be possible to carry on intelligible conversation between the two transceivers by talking in a normal tone of voice at a distance of not less than one foot from each speaker. See paragraph 4.19.

3.47 Jag.- Jag shall be measured by examining the spots of the photographic copy received on Royal Bromide paper, or equal, with a maximum level of plus 2 db. See paragraph 4.20.

3.48 Line Feed.- Line feed should be measured on a receiving machine recording or Royal Bromide paper, or equal, at a maximum level of + 2 db. See paragraph 4.21.

3.49 Dielectric Strength Test.-

3.49.1 Synchronous Motor Coils.- Before and after installation in the end bell of the motor, the synchronous motor coils shall be tested for shorter tones as specified in paragraph 4.22.1.

3.49.2 Stator of Start Motor.-

3.49.2.1 A dielectric strength test shall be performed between windings, and between windings and ground both before and after compression of the coils on the stator is completed. See paragraph 4.22.2.

3.49.2.2 A dielectric strength test shall be performed between coils and frame of motor after stator is mounted in motor frame and winding terminals are soldered. See paragraph 4.22.2.

3.50 Motor Power Output.-

3.50.1 Partially Assembled Motors.- The motor, with countershaft and worm gear removed, shall be tested to determine the shock torque and steady state torque. The steady state torque shall be at least 9 ounce-inches with a motor input current of 65 milliamperes. The shock (suddenly applied) torque shall be at least 5 ounce-inches with a motor input current of 65 milliamperes. See paragraph 4.23.1.

3.50.2 Assembled Motors.- Failure in more than 10 of 50 attempts shall constitute failure and result in rejection of the motor. See paragraph 4.23.2.

3.50.3 Automatic Stop Operation.- Motor should stop as shown in paragraph 4.23.3.

3.51 Motor Inspection.-

3.51.1 Life Test of Motor.- There shall be no evidence of excessive wear, overheated coils or loose particles of foreign material within the gear box or motor casing. See paragraph 4.24.1.

3.51.2 Alternate Speed Test.- The received copy shall compare favorably with copy provided by two approved samples operating together; there shall be no evidence of irregular skew and the uniform skew shall not exceed 0.070 inch in 12 inches of copy. See paragraph 4.24.2.

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3.51.3 External Oscillator Operation.- The received copy shall be at least as free from recording defects as copy recorded by approved samples operating together at a down speed of 33 1/3 rpm.

3.52 Service conditions.-

3.52.1 Equipment operating.- The equipment shall meet the requirements of this specification under any probable combination of the following:

(a) Temperature.- Any ambient temperature in the range of -40° to +131°F.

(b) Elevation.- Any elevation up to 10,000 feet above sea level.

(c) Relative humidity.- Any relative humidity up to 98 percent.

3.52.2 Equipment non-operating.- The equipment shall meet the requirement of 3.34.1 after subjection to any one or more of the following non-operating conditions:

(a) Temperature.- Continuous exposure for 72 hours at 160°F and 72 hours at -80°F at any orientation.

(b) Elevation.- Any elevation up to 50,000 feet above sea level.

(c) Relative humidity.- Relative humidity up to 100 percent including condensation caused by temperature changes.

3.53 Vibration.-

3.53.1 Vibration.- Except for internal resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second when shock mounts (if any) are blocked or removed.

3.54 Preproduction samples.- The contractor shall furnish preproduction samples for approval as required by the invitation for bids and contract. See paragraph 6.2 (e) (1).

3.55 Interchangeability.- Corresponding units and replaceable assemblies, subassemblies and parts on contract shall be physically and functionally interchangeable as complete items without modification thereof or of other articles with which the units are used. (See 4.5). When dimensions, ratings, characteristics, etc, are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. If the contractor is in doubt as to whether a particular assembly, subassembly or part is to be considered replaceable, the contracting officer shall be consulted.

3.56 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2).

3.57 Exhaust Filter Assembly.- To be constructed as shown in Drawing SC-DL-201677.

3.58 Table MT-252( )/TXC-1.- To be constructed as shown in Drawing SC-DL-37476.

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3.59 Workmanship.- The equipment shall be manufactured and assembled in a throughly workmanlike manner, in accordance with the applicable portions of the following paragraphs herein:

3.1 Construction

3.2 Brazing, soldering, welding and cleaning

3.3 Finishes

3.5 Tropicalization

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.2.)

(b) Acceptance inspection of items before they are prepared for delivery. (See 4.3.)

(c) Acceptance inspection of preparation for delivery. (See 4.4 and Section 5.)

4.2 Preproduction inspection.- This inspection shall consist of the group A inspection and nondestructive group B inspection specified in tables II and III, respectively, and the preproduction inspection specified in table I. Other nondestructive inspection on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I - Preproduction inspection

Inspection (For additional preproduction inspection see 4.2)	Reqt Par.	Test Par.
Temperature	3.52	4.7.2
Moisture Resistance	3.52	4.7.3
Vibration	3.53	4.7.1

4.3 Acceptance inspection (see 4.1.2(b)).- This inspection shall include group A and group B, inspection as specified in 4.3.1 through 4.3.2.3. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.3.1 Group A inspection.- This inspection (including sampling) shall conform to table II and Standard MIL-STD-105. It shall be performed in any order which is satisfactory to the Government inspector, except that the operational inspection shall be last.

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Table II - Group A Inspection

Inspection	Reqt Para	Insp Para	AQL
Visual and mechanical, Major		4.6	4.0 dphu
Visual and mechanical, Minor		4.6	15.0 dphu
Synchronizing frequency	3.36	4.9	4.0 dphu for group combined
Phasing	3.38	4.11	
Positive	3.38.1	4.11.1	
Negative	3.38.2	4.11.2	
Motor running	3.38.3	4.11.3	
Pick-up from local frequency	3.39	4.12	
Fork beat	3.40	4.13	
Output level	3.42	4.15	
Contrast	3.43	4.16	
Regulation	3.44	4.17	
Talkback circuit	3.46	4.19	
Jag	3.47	4.20	
Linefeed	3.48	4.21	
Motor test	3.45	4.18	
Acceleration	3.45.1	4.18.1	
Motor current	3.45.2	4.18.2	
Coasting	3.45.3	4.18.3	
Clutch tension	3.45.4	4.18.4	
Motor power output	3.50	4.23	
Assembled motors synchronization	3.50.2	4.23.2	
Automatic stop	3.50.3	4.23.3	
Motor inspection and test	3.51	4.24	
Life test of motor	3.51.1	4.24.1	
Alternate speed test (30 rpm)	3.51.2	4.24.1	
External oscillator operation (33 rpm)	3.51.3	4.24.3	
Motor power output, partially assembled motors	3.50.1	4.23.1	1.0%
Operational	3.37	4.10	1.0% AQL for group combined
Recording	3.37.1	4.10.1	
Photographic	3.37.2	4.10.2	
Direct recording	3.37.3	4.10.3	
Operation	3.37.4	4.10.4	
Spot size	3.37.5	4.10.5	
Microphonic tubes			
Transmission	3.35	4.8	

All electrical effects will be considered major.

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4.3.2 Group B inspection.- This inspection (including sampling) shall conform to table III and the Appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. Disposition of rejected product (sample units and inspection lots) shall be in accordance with Standard MIL-STD-105 and 4.3.3.

4.3.2.1 Group B sampling plans.- The group B sampling plans, as listed in table III, shall be as follows:

Group B plan	AQL	Inspection level for normal inspection	Inspection level for reduced inspection
B-1	4.0%	L-8	L-6
B-2	6.5%	L-7	L-5

4.3.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.3.2.3 Procedure in case of failure.- When an inspection lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.3.3 Disposition of nonconforming product.- When defective sample units or rejected inspection lots are resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

Table III - Group B Inspection

Inspection	Reqt Para	Insp Para	Sampling Plan
Receiver sensitivity	3.41	4.14	
600 ohm termination	3.41.1	4.14.1)	B-2 for
2500 ohm termination	3.41.2	4.14.2)	combined group
Dielectric strength	3.49	4.22	
Synchronous motor coils	3.49.1	4.22.1	B-1
Stator of start motor	3.49.2	4.22.2	B-1

4.4 Acceptance inspection of preparation for delivery.-

4.4.1 Preservation and packaging.- Inspection of preservation and packaging shall be as specified in Specification MIL-P-116. Classification of defects shall conform to Table IV and V.

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4.4.2 Packing and marking of exterior containers.- Packing and marking of exterior containers shall be given visual inspection for the defects listed in Table IV and V and to determine conformance with the approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the Appendix to Standard MIL-STD-105. Inspection Level L-8 shall be used for normal inspection and L-6 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL for major defects shall be four percent and the AQL for minor defects shall be ten percent.

4.4.2.1 Inspection lot.- A lot for visual inspection of the pack shall be all completed packs which are identical and are submitted for inspection at one time.

4.4.2.2 Procedure in case of failure.- If an inspection lot is rejected the contractor shall immediately investigate the cause of failure and shall report to the Government Inspector the results thereof and details of the corrective action taken. If the contractor and Government Inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the Contracting Officer for resolution.

4.4.2.3 Disposition of nonconforming product.- Disposition of rejected lots and defective sample units shall be in accordance with Standard MIL-STD-105. When resubmitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.5 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.55. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

TABLE IV

CLASSIFICATION OF DEFECTS FOR PRESERVATION, PACKAGING & MARKING

MAJOR DEFECTS

1. Use of improper or defective material.
2. Quantity in unit package not as specified.
3. Incorrect packaging method applied.
4. Cushioning or padding omitted.
5. Cushioning or padding inadequate for the protection of the barrier material from projections, sharp edges, or other similar features of the item.
6. Cushioning inadequate for the physical and mechanical protection of the item.
7. Desiccant omitted.
8. Desiccant not in proper quantity to meet minimum requirements.
9. Unsealed, punctured, or imporperly sealed barrier bag, wrap, or envelope.
10. Stock number omitted, incorrect, or illegible.
11. Nomenclature omitted, incorrect, or illegible.
12. Marking of quantity of items in package omitted, incorrect, or illegible.
13. Different stock numbered items in the same unit package.

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MINOR DEFECTS

1. Item not properly blocked or braced within the unit package to prevent movement.
2. Humidity indicator omitted.
3. Packaging material damaged.
4. Corners of inner carton not blunted (Method II only).
5. Air not expelled from barrier.
6. Desiccant not properly secured.
7. Insufficient area of barrier material provided to permit 2 subsequent seals.
8. Any item of marking information other than 10 thru 12 omitted, incorrect, or illegible.

TABLE V

CLASSIFICATION OF DEFECTS FOR PACKING & MARKING

MAJOR DEFECTS

1. Use of improper or defective material.
2. Quantity in pack not as specified.
3. Gross weight in excess of specified amount.
4. Box closure not as specified.
5. Type, grade, class, and style of the shipping container not as specified.
6. Strapping omitted.
7. Strapping inadequate or incorrectly applied.
8. Items not adequately blocked, braced, or cushioned within the shipping container to prevent movement or damage.
9. Shipping documents or packing list omitted.
10. Stock number omitted, incorrect, or illegible.
11. Nomenclature omitted, incorrect, or illegible.
12. Marking of quantity of items in pack omitted, incorrect, or illegible.
13. Destination marking omitted, incorrect, or illegible.
14. Service designation (color marking) omitted.
15. Set or assembly markings omitted.
16. Specified special marking and labeling such as MAP labels, shipment digit markings, etc., omitted, incorrect, or illegible.
17. Overseas code marking omitted, incorrect, or illegible.

MINOR DEFECTS

1. Unsealed carton.
2. Defective taping or sealing of carton.
3. Any other box defect which may be considered minor by definition of Standard MIL-STD-105.
4. Any item of required marking information other than 10 thru 17 listed under major defects omitted, incorrect, or illegible.

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4.6 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and table VI.

Table VI - Classification of Visual and Mechanical Defects

Classification	Defect
Major	Improper lead dress or cabling or wires Insulation burned, abrased, pinched. Taut wire Insulation frayed
Minor	Broken strands Insulation back more than 1/8 inch

4.7 Service conditions tests.-

4.7.1 Vibration test.- The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.53 as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration. (See 4.7(d)).

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because the unit weighs more than 250 pounds or because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

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4.7.2 Extreme Temperature Test.- The equipment shall be subjected to the following extreme ambient dry temperature test to determine compliance with service requirements of Section 3. It will be considered that stabilization of the equipment's temperature has been reached when tree consecutive temperature readings (at 10-minute intervals) taken at suitable points within the equipment show no more than a ± 2 C differential.

(a) Start at room temperature (21 to 32 C); take electrical measurements to establish a standard of performance. The units shall be checked to insure compliance with the requirements of paragraph E-2g, and to insure a satisfactory transmitting signal as viewed on an oscilloscope.

(b) Remove back cover and motor cover, and raise the temperature to 49 ± 3 C. Allow the equipment to remain at this temperature for three hours.

(c) Replace the back cover and motor cover and maintain at 49 ± 3 C for three hours operating the equipment; repeat measurements as in (1) during the last hour.

(d) Reduce to room temperature and stabilize; repeat measurements as in (1).

(e) Remove back cover and motor cover and reduce to 0 C. Allow the equipment to remain at this temperature for three hours. Replace the covers and after 20 minutes operation repeat measurements as in (1).

(f) Raise temperature to 25 C and stabilize; repeat measurements as in (1).

4.7.3 Relative humidity.- The following test shall be conducted to insure that the equipment will be capable of providing satisfactory performance in field service during and after exposure to tropical conditions.

4.7.3.1 Test Procedure.- The equipment shall be tested as follows:

(a) Measure the performance (in accordance with the production tests as specified in paragraphs 3.52 of the equipment. Realigon or readjust if necessary to meet specification requirements.

(b) Subject the equipment to continuous cycling for a total of 5 of the 48-hour cycles shown on MIL-STD-170(SigC), and measure performance in accordance with the following during the periods indicated on that drawing:

(1) Synchronizing Frequency (4.9)

(2) Contrast (4.16)

(3) Binding Post Output.- Measure in accordance with 4.15.

(4) Phasing.- Turn transceiver SELECTOR switch to RECORD PHOTO. With no input signal, turn GAIN control to 100. Start motor, and while depressing PHASING button, sharply tap the chassis with a screwdriver handle. The phasing magnet should be activated by the tapping and should release the clutch stop arm. Visually inspect the exposed clutch mechanism for excessive corrosion or sticking parts.

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4.7.3.1 Test Procedure(contd).-

(5) Transmitting Output.- View output of the equipment on oscilloscope (similar to manner of viewing a transmitter's output.)

(c) Do not remove the equipment from the test chamber for measurements.

(d) Start above measurements not more than 1 hour after power has been applied to the equipment. Complete above measurements as rapidly as possible. Do not leave power on the equipment after measurements have been completed. Perform remainder of production tests as soon as possible.

(e) At the termination of the 5 humidity cycles, repeat the tests of paragraphs 4.9 thru 4.19 except 4.13.

4.7.3.2 No realignment and/or readjustment of the system may be made during the test, exclusive of the initial adjustments (4.7.3.2) and normal operating controls. If any component fails to meet the performance required, it shall be considered as failing to pass the test.

4.8 Transmission.- Levels shall be set for negative transmission with input power of 115 volts, 50 cycles per second. During the transmission of the entire copy, the frequency shall be continuously varied from 45 to 55 cycles per second and back to 45 cycles per second at the approximate rate of once every 20 seconds. The voltage during the transmission of the first quarter of the copy shall be 100 volts; during the second quarter 110 volts; during the third quarter 120 volts; during the fourth quarter 130 volts. (See para 3.35)

4.8.1 Levels shall be set for negative transmission with input power at 115 volts, 65 cycles per second. During the transmission of the entire copy the frequency shall be continuously varied from 60 to 70 cycles per second and back to 60 cycles per second at the approximate rate of once every 20 seconds. The voltages maintained during transmission shall be as specified in paragraph 4.8 .

4.8.2 The receiving unit shall be set for proper recording level with a power input of 115 volts, 50 cycles per second. During the reception of the entire copy the frequency shall be continuously varied from 45 to 55 cycles per second and back to 45 cycles per second at the approximate rate of once every 20 seconds. Each quarter of the copy shall be received with a different input voltage as follows: 100, 110, 120, and 130 volts, respectively.

4.8.2.1 The receiving unit shall be set for proper recording level with a power input of 115 volts, 65 cycles per second. During the reception of the entire copy the frequency shall be continuously varied from 60 to 70 cycles per second and back to 60 cycles per second at the approximate rate of once every 20 second. The voltages maintained during reception shall be as specified in paragraph 4.8.2.

4.9 Synchronizing Frequency.- A standard signal source of 1800 cycles per second shall be used as the reference frequency. The frequency of the fork oscillator in each facsimile transceiver on order shall be compared with the standard frequency after the power (with the line voltage at 130 volts) has been turned on the unit under test for periods of 30 minutes and 2 hours respectively. (See para 3.36)

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4.10 Recording.-

4.10.1 Photographic.- (A signal of suitable level shall be generated by an approved sample transceiver. Each transceiver on order shall be set up for photographic reception. The transmitted signals shall be recorded in strips at least 1/2-inch wide on Royal Bromide Paper, P-1, or equal at each of the following levels on the meter of the receiving machine:- 2, 0, +2, and +3. The strip corresponding to +2 dbm level shall be dark gray. "A dark gray is defined as a reflectance of 13 percent or less, as measured with a photo-electric reflection meter, Photovolt Model 610 (as made by Photovolt Corporation, New York, N.Y.)., or equal. The development of the paper shall have been performed in fresh Developer D-72, diluted with an equal volume of water, at a temperature of 68F and agitated intermittently for one and one-half minutes. Recordings made at ≠ 3 db shall be darker than readings made at ≠ 2db.")

4.10.2 Direct Recording.- (The above procedure shall be repeated using Teledeltos Grade H for recording at levels of -1, 0, +1, +2, and +3. The strip corresponding to +2 dbm meter reading shall be dark gray. "For Teledeltos recording, a dark gray is defined as a reflectance of 10.3 per cent, or less, as measured with a photo-electric reflection meter, Photovolt Model 610, or equal. Recordings made at ≠ 3 db shall be darker than readings made at ≠ 2 db,")

4.10.3 Test of Operation.- Each transceiver on order shall be tested for proper operation as follows: (See para 3.37.3)

a. Negative reception on Royal Bromide Paper, P-1, or equal.
b. Positive reception on Teledeltos, Grad H, or equal.
c. Negative transmission

4.10.4 Spot Size.- The copy shall be examined under a microscope having a graduated scale in the field of view to determine the size of the spot, or shall be compared, using a magnifying glass, to standard test copy illustrating satisfactory limits of spot size. The measurements shall be made on Royal Bromide F1 Paper or equal at a level of ≠2 db on the meter of the receiving machine. (See par 3.37.4)

4.10.5 Microphonic Tubes.- Each unit shall be tested as follows:

4.10.5.1 With transceiver operating from the rectifier power unit, selector switch on RECORD PHOTO, GAIN control at 100, and nothing connected to the signal input terminals, push the phasing button.

4.11 Phasing.- Each transceiver shall be tested as follows:

4.11.1 Positive.- With the machine set at RECORD PHOTO and operating from an a-c power source at 100 volts, a positive phasing signal of 8 db contrast shall be applied to the low level input terminals. (See par 3.38.1)

4.11.2 Negative.- The above test shall be repeated using negative phasing signals of 6 db contrast. (See par 3.38.2)

4.11.3 With motor running and the phasing button released and with dog latch at a position where it must lose nearly one revolution before engaging with the stop arm, the drum and leadscrew assembly shall decelerate until the dog latch engages the stop arm of the clutch assembly. (See par 3.38.3)

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4.12 Pick-Up from Local Frequency Generator.- Each transceiver shall be set up as follows: The selector switch shall be set at RECORD PHOTO, the GAIN control shall be set at 100, and there shall be no signal input. (See paragraph 3.39.)

4.13 Fork Beat.- With motor running an 1800.5 cycle per second signal shall be introduced into the transceiver at the 600 ohm binding posts. The level of this signal shall be such that with the GAIN control at 100 the meter in the transceiver reads approximately -10. This test shall be repeated with the same requirements inserting the test signal at the miniature female connector and again inserting the test signal at the RADIO RCVR jack. (See paragraph 3.40.)

4.14 Receiver Sensitivity.-

4.14.1 600 Ohm Termination.- A steady 0-dbm signal from an approved sample transceiver set for TRANSMIT, shall be passed through a calibrated attenuator pad and into the miniature connector on each transceiver on order. (See paragraph 3.14.1.)

4.14.2 2500 Ohm Termination.- When a 0.25 volt, 1800 cycle per second signal is applied at the RADIO RCVR jack, and also when a 12 volt, 1800 cycle per second signal is similarly applied. (See paragraph 3.14.2.)

4.15 Output Level Test.- Each of the terminations listed below shall be connected in turn to the proper terminals of each facsimile transceiver. In each case the transceiver shall be adjusted to give a +2 dbm reading on the transceiver meter. A known attenuator pad and the db meter with an accuracy of 3 percent of 1800 cu per second shall be connected across the termination and measurements made. (See paragraph 3.42.)

4.15.1 Limits.-

Output	Termination
Miniature Connector	600-ohms
Binding Post	600-ohms
Transmit Jack	100-ohms
*UC Coil (3 ohm)	open circuit
*UC coil (15 ohm)	open circuit

*Input to coil adjusted to 15.46 volts.

4.16 Contrast.- The difference between maximum and minimum signals set up on the phasing ring shall be measured with each unit set for both positive and negative transmission. The signals shall be obtained from the black and white parts of the phasing ring, respectively. The measurement shall be made using a suitable attenuator pad and db meter specified in paragraph 4.15. (See paragraph 3.43.)

4.17 Regulation.- Each transceiver shall be tested as described herein. A variable transformer such as a variac shall be connected in the power input and normal frequency shall be used. The following test shall be conducted:

a. Adjust input voltage of the transceiver to 115 volts a.c. (RMS).
b. Set contrast control for a minimum reading of the transceiver meter on the black portion of the phasing ring.

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4.17 Regulation.- (Cont'd)

c. Turn the cylinder to the white portion of the phasing ring.
d. Adjust gain control until meter reads 0 db.
e. Change the input to 130 volts a.c. (RMS) for 15 seconds and note the meter reading.
f. Change the input to 100 volts a.c. (RMS) for 15 seconds.
g. Vary the input uniformly between limits of 100 - 130 volts a.c. (RMS). Each complete cycle of frequency change (100 - 130 - 100 volts) shall be accomplished in approximately 1 second. (See paragraph 3.44.)

4.18 Motor Test.- Each transceiver shall be set up as specified in paragraph 4.17 for regulation. The selector switch shall be set at RECORD PHOTO and the a-c power input adjusted to 100 volts. When the START button is depressed and held, the cylinder shall start turning and accelerate to a speed greater than 60 rpm. Upon releasing the button, the drum speed shall drop to 60 rpm and the drum shall continue to rotate at that speed. (See paragraph 3.45.1)

4.18.1 Motor acceleration.- With motor fuse removed and with transceiver set up as specified in 4.17, the selector switch shall be set at "RECORD PHOTO" and input at 130 volts. The drum shall be at the extreme right hand end of leadscrew with half-nuts engaged. The start button shall be pressed until motor reaches maximum speed and then shall be released. (See paragraph 3.45.2.)

4.18.2 Motor current.- Start motor, tune selector switch to "RECORD PHOTO" position. Motor current potentiometer (R-93) should adjust motor current between 65 and 75 M.A. Set motor current at 75 M.A. Turn selector switch to "TRANSMIT" position. (See paragraph 3.45.2.)

4.18.3 Motor coasting.- Set line voltage to 130 volts. Remove 1/8 ampere fuse. Move drum to extreme right hand side of lead screw. Engage split nuts. Hold key switch in "START" position until motor accelerates to full speed. Release key switch. (See paragraph 3.45.3.)

4.18.4 Clutch tension test.- (See paragraph 3.45.3.)

(a) Hook loop of tension scale onto edge of slot on phasing ring.
(b) Suspend clutch tension scale.
(c) Allow drum to rotate until string of clutch tension is taut. Read primary clutch tension. (Limit 8 to 13 ounces)
(d) Allow stop arm to rest against phase magnet amature.
(e) Read secondary clutch tension. (Limit 3 to 6 ounces).

4.19 Talkback Circuit.- A signal of suitable level shall be generated by an approved sample transceiver with the selector switch in TRANSMIT position. A circuit shall be established between the sample and the transceiver under test. The transceiver on test shall be adjusted for +2 db meter reading in RECORD PHOTO position. The selector switch of both transceivers shall be turned to STAND BY position and the talkback circuits shall be checked.

4.20 Jag.- Jag between adjacent scanning lines of the received copy shall be considered acceptable even though recurring throughout the picture, provided the displacement of any of the spots is no greater than 0.005 inch from either of the corresponding spots contained in adjacent scanning lines. Jag shall only be measured on the receiving equipment. (See paragraph 3.47.)

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4.21 Line Feed.- Line feed shall be measured as a relative displacement between adjacent scanning lines. Line feed causing no more than 0.001 inch variation in spacing between adjacent scanning lines shall be acceptable.

4.22 Dielectric Strength test.-

4.22.1 Synchronous Motor Coils.- The synchronous motor coils shall be tested for shorted turns by appling 850 volts r.m.s., 12 kc a.c., for 10 seconds, between the coils and ground. (See paragraph 3.49.1.)

4.22.2 Stator of start motor.- A dielectric strength test shall be performed on the stator coils of the Start Motor with a potential of 1000 volts r.m.s., 60 cycle a.c. applied for one minute. (See paragraph 3.49.2.)

4.23 Motor Power Output.-

4.23.1 Partially Assembled Motors.- With the motor energized from a circuit simulating the motor amplifier circuit of Facsimile Set AN/TXC-1( ) operating a 117 volts, 60 cps, and with 115 volt 60 cps start winding completely disconnected, the steady state torque shall be measured. (See paragraph 3.50.1.) The brake used for determining the torque shall be a non-ferrous metal disc mounted on the motor and loaded at its periphery by the magnetic field of an electro-magnet which transmits the torque through an arm to a scale or calibrated weight. The torque shall be computed by obtaining the product of the force acting on the disc by the distance from the center of the disc to the point on the disc where the magnetic field is applied. The current in the coils of the electro-magnet shall reach 63% of the steady state value within 1/3000 second after the voltage is applied to the coils.

4.23.2 Assembled Motors.- The motor and drum shall be started with split nuts disengaged and with phasing armature held against the phasing magnet. With a motor input current of 70 ma. and with a line voltage of 117 volts, the motor shall lock into synchronism. (See paragraph 3.50.2.)

4.23.3 Automatic stop.-

(a) Turn selector switch to "RECORD PHOTO" position.

(b) Connect 1800 cycle standard signal to "LINE JACK."

(c) Adjust "GAIN" control for +2 db reading.

(d) Disconnect recorder lamp.

(e) Insert recorder lamp into socket of adapter and plug adapter into recorder lamp socket.

(f) Connect milliameter to connections of adapter.

(g) Disengage half nuts.

(h) Move the drum to the left so that about 1/4 inch of thread of lead screw is visible.

(i) Start motor and engage half nuts.

(j) Allow drum to rotate until limit switch operates.

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4.23.3 Automatic stop (contd).-

(k) Motor should stop and drum should stop rotating and current of recorder lamp should drop to 0.

(l) Disengage half nuts.

(m) Move drum to right to release limit switch.

(n) Motor should remain stopped and current of recorder lamp should remain at 0.

(o) Press "START PHASE" switch to "START" and then release switch.

(p) The synchronous motor should not operate.

(q) Turn selectorswitch to "STANDBY" position.

(r) Start motor.

(s) Synchronous motor should operate.

(t) Turn selector switch to "TRANSMIT" position.

(u) Repeat steps (g), (h), (i), (j).

(v) Motor should stop and drum should stop rotating.

(w) Repeat steps (l), (m).

(x) Motor should remain stopped.

(y) Repeat steps (o), (p), (q), (r), (s).

The paper shall not burn at the end of any cycle.

4.24 Motor Inspection and Test.-

4.24.1 Life Test of Motor.- The motor shall be operated for at least 1000 hours at full load, either in Facsimile Transceiver TT-1( )/TXC-1 or on a test block and powered by a circuit simulating the motor drive circuit in Facsimile Transceiver TT-1( )/TXC-1. The motor shall be loaded by blocking the dog arm on the clutch assembly and adjusting the primary clutch until the slipping torque is between 36 and 42 ounce-inches. At the completion of the life test, the motor shall be disassembled and inspected. (See para 3.51.1)

4.24.2 Alternate Speed Test.- The speed of the drum in Facsimile Transceiver TT-1( )/TXC-1 shall be tested with its speed switch set at 1000 cps. At this setting, the drum shall rotate at a speed of 33-1/3 rpm. The equipment under test shall be adjusted for positive transmission on photographic paper, with its speed switch set at the 1000 cps position. Another Facsimile Transceiver TT-1( )/TXC-1 shall be connected to the transceiver under test and adjusted to transmit positively with its speed control set at the 1000 cps position. A copy shall be received by the transceiver under test with line voltage varied in accordance with paragraph 4.8.2. It is not necessary to vary the line frequency for this test. (See para 3.51.2)

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4.24.3 External oscillator operation test.- A representative recording (not less than 1 inch by 10 inches) on photographic paper shall be made from another transceiver operating as a transmitter at 33-1/3 rpm. A test pattern made of black lines approximately 0.015 inch wide on a white background shall be placed on the drum of the transmitting machine with the lines on the test pattern parallel to the axis of the lead screw. For this test a source of 100 cps derived from a secondary frequency standard may be used at each transceiver, or a common source of 1000 + 1 cps may be used for both transceivers. (See para 3.51.3)

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- The components of Facsimile Set AN/TXC-1( ) shall be preserved, packaged and tested in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described herein.

5.1.1.1 Technical manual.- Each technical manual shall be packaged Method IC-3, as follows: Inclose each technical manual within a close-fitting, Type II, Class b bag, conforming to Specification MIL-B-117. Heat seal bag to effect closure.

5.1.1.2 Tubes.- Remove all operating tubes over four inches in length not securely clamped in place and package together with spare tubes, individually in accordance with the applicable provisions of Specification MIL-P-75.

5.1.1.3 Cords.- Each cord shall be packaged individually Method III as follows: Wrap each cord under four feet in length in flexible, single face, corrugated fiberboard, Type I, conforming to Specification LLL-F-291. Secure the wrap with gummed paper tape, Class 2, conforming to Specification UU-T-111. Wind each cord over four feet in length into a coil of proportionate dimensions. Tie at three places with cotton twine, Type II, 8 ply, conforming to Specification T-T-871. Secure connectors to the coil with the ties. Wrap coiled cords within flexible, single face, corrugated fiberboard, Type I, conforming to Specification LLL-F-291. Secure the wrap with gummed paper tape, Class 1, conforming to Specification UU-T-111.

5.1.1.4 Screwdriver, split units.- Each item shall be packaged individually, Method III as follows: Wrap each item with flexible, single face, corrugated fiberboard, Type I, conforming to Specification LLL-F-291. Secure the wrap with gummed paper tape, Class 1, conforming to Specification UU-T-111.

5.1.1.5 Stylus.- Each stylus shall be packaged individually Method III as follows: Cushion each stylus with flexible, single face, corrugated fiberboard, conforming to Specification LLL-F-291. Secure cushioning with gummed paper tape, Class 1, conforming to Specification UU-T-111. Place each cushioned stylus within a close-fitting paper bag, conforming to Specification UU-B-36. Seal closure with Class 1, gummed paper tape, conforming to Specification UU-T-111.

5.1.1.6 Intermediate container.- Consolidate the stylus, packaged as specified in 5.1.1.5, within a close-fitting folding box, Style II, Type A, Class 2, conforming to Specification PPP-B-566. Closure shall be made in accordance with the appendix of the box specification.

5.1.1.7 Scale.- Each scale shall be packaged Method IA-8 as follows: Cushion each scale by wrapping in flexible, single face, corrugated fiberboard, Type I, conforming to Specification LLL-F-291. Secure the wrap with gummed paper tape, Class 2, conforming to Specification UU-T-111. Place each cushioned scale within a close-fitting, bag, made of water vaporproof barrier material qualified for Class 2 application conforming to Specification MIL-B-131 and fabricated in accordance with Specification MIL-B-117. Heat seal to effect closure.

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5.1.1.8 Cover.- Each cover shall be packaged Method III as follows: Fold each canvas dust cover into a bundle of proportionate dimensions and tie with cotton twine, Type II, 8 ply, conforming to Specification T-T-871.

5.1.1.9 Consolidated package.- Consolidate the items, packaged as specified in 5.1.1.2 through 5.1.1.8, together within a close-fitting, Type I, Class 2, Grade 5, box, conforming to Specification PPP-B-636. Closure shall be in accordance with the appendix of the box specification. In addition, seal all reams and joints with waterproof, pressure sensitive tape, Type III, Class 3, any color, conforming to Specification PPP-T-60.

5.1.1.10 Facsimile Transceiver TT-1( )/TXC-1( ).- Each transceiver shall be packaged Method IIb as follows: Cushion each transceiver on all surfaces with cells or pads or both, fabricated of fiberboard, having a minimum dry bursting strength of 200 pounds, designed to absorb the shock of impact, encountered in handling and transit. Place each cushioned transceiver together with the required number of units of activated desiccant within a close-fitting RSC, Type I, Class 1, box, conforming to Table I of Specification PPP-B-636. The desiccant shall conform to Specification MIL-D-3464, in the quantity required by the formula in Specification MIL-P-116. The desiccant shall be placed and secured within the cushioning cells. Seal the entire box closure with gummed paper tap, Class 2, conforming to Specification UU-T-111. The box corners shall be blunted and any stapled or stitched joints shall be covered with gummed paper tape. The sealed interior box shall be placed within a water vaporproof barrier, conforming to Specification MIL-B-131, Class 1, The barrier material shall be cut from a bag and fabricated with one dimension to allow three resealings for inspection purposes. A humidity indicator and, conforming to Standard MS-20003, shall be placed within the barrier on top of the sealed interior box and secured with pressure sensitive tape, any type, class or color, conforming to Specification PPP-T-60. Heat seal the water vaporproof barrier bag, leaving a small opening to exhaust the air. Evacuate the excess air and complete the heat seal. The time lapse between positioning of the desiccant and the completion of the barrier bag seal must be held to an absolute minimum. Place the water vaporproofed transceiver within a close-fitting, Type I, Class 2, Grade 5, box, conforming to Specification PPP-B-636. Seal the box closure in accordance with the appendix of the box specification.

5.1.1.11 Paper; recording.- Paper shall be packaged, following procedures specified in 5.1.1.1.

5.1.1.12 Rectifier Power Unit PP-86( )/TXC-1.-

5.1.1.12.1 Tubes, fuses and cover.- Each item shall be packaged as specified in 5.1.1.2, 5.1.1.5 and 5.1.1.8 respectively.

5.1.1.12.2 Consolidated package.- Consolidate the items packaged as specified in 5.1.1.11 and 5.1.1.12.1 together as specified in 5.1.1.9.

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5.1.1.12.3 Rectifier Power Unit PP-86( )/TXC-1.- Each power unit shall be packaged Method IIb as follows: Wind each cord into a coil of proportionate dimensions. Tie at three places with cotton twine securing the connectors to the coil with the ties. Secure each coiled cord to the unit with cotton twine and continual packaging as specified in 5.1.1.10.

5.1.1.13 Table.- Each table shall be packaged Method IC-5 as follows: Collapse legs of table. Cushion each table on all surfaces with cells or pads or both, fabricated of fiberboard, having a minimum dry bursting strength of 200 pounds, and designed to absorb the shock of impact, encountered in handling and transit. Cells and pads shall have a water resistance equal to that of the box in which they are used. Place each cushioned repeater within a close-fitting, Type I, Class 2, Grade 5, box, conforming to Specification PPP-B-636. Box closure shall be in accordance with the appendix of the box specification. In addition all seams and joints shall be sealed with waterproof, pressure sensitive tape, Type III, Class 3 any color, conforming to Specification PPP-T-60.

5.1.2 Level C.- Components of Facsimile Set AN/TXC-1( ) shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage, during direct shipment to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.-

5.2.1.1 Transceiver TT-1( )/TXC-1( ).- Each transceiver packaged as specified in 5.1.1.10 and consolidated package as specified in 5.1.1.9 shall be packaged together within a close-fitting Style 4, Class 2, sealed, wood box, conforming to Specification PPP-B-621. Box closure shall be in accordance with the box specification.

5.2.1.2 Rectifier Power Unit PP-86( )/TXC-1( ).- Each rectifier power unit packaged as specified in 5.1.1.12 and consolidated package specified in 5.1.1.1.2.2 shall be packed together for shipment as specified in 5.2.1.1.

5.2.1.3 Table MT-252/TXC-1( ).- Each table packaged as specified in 5.1.1.1.3 shall be packed for shipment as specified in 5.2.1.1

5.2.1.4 Technical manual.- Place the packaged technical manual between the contents and the lid of the number 1 box.

5.2.1.5 Boxes containing components of Facsimile Set AN/TXC-1( ) packed as specified above, shall be arranged in the following sequence:

Facsimile Transceiver TT-1( )/TXC-1 Consolidated Package-Technical Manual	Box No. 1/3
Rectifier Power Unit PP-86( )/TXC-1 Consolidated Package	Box No. 2/3
Table MT-252( )/TXC-1	Box No. 3/3

5.2.1.6 Metal strapping.- Shipping containers shall be strapped with flat steel strapping, in conformance with the requirements of the appendix of the container specification, only for direct shipment to ports.

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5.2.2 Level B.- Facsimile Set AN/TXC-1( ) shall be packed for shipment as specified in 5.2.1, except that the nailed wood boxes shall conform to Style 5, Class 1, of Specification PPP-B-621.

5.2.3 Level C.- Facsimile Set AN/TXC-1( ) shall be packed for shipment in a manner, conforming to the requirements of Uniform Freight Classification for rail shipment, National Motor Freight Classification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.3 Marking.- Interior packages and exteior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Facsimile Set AN/TXC-1( ) is used to transmit and receive written, printed or pictorial matter over wire or short radio communications facilities.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and data of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction samples required:

(1) Two of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample. (See 3.54)

(2) Preproduction pack(s) as follows:

(1) Makeup of pack(s).
(2) Number of each king of pack to be submitted.
(3) Inspection to be performed thereon--including rough handling test, which will not be performed as acceptance inspection.
(4) Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. This data should be submitted to the contracting officer with the preproduction pack.
(5) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

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6.2 Ordering data.- (Cont'd)

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required. (See 3.56.)

(i) Quantity of tools and running spare parts required. (See 3.56.)

(j) Submission of the statement of treatment referenced in 3.52.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/TXC-1W. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information. (See 1.1.)

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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